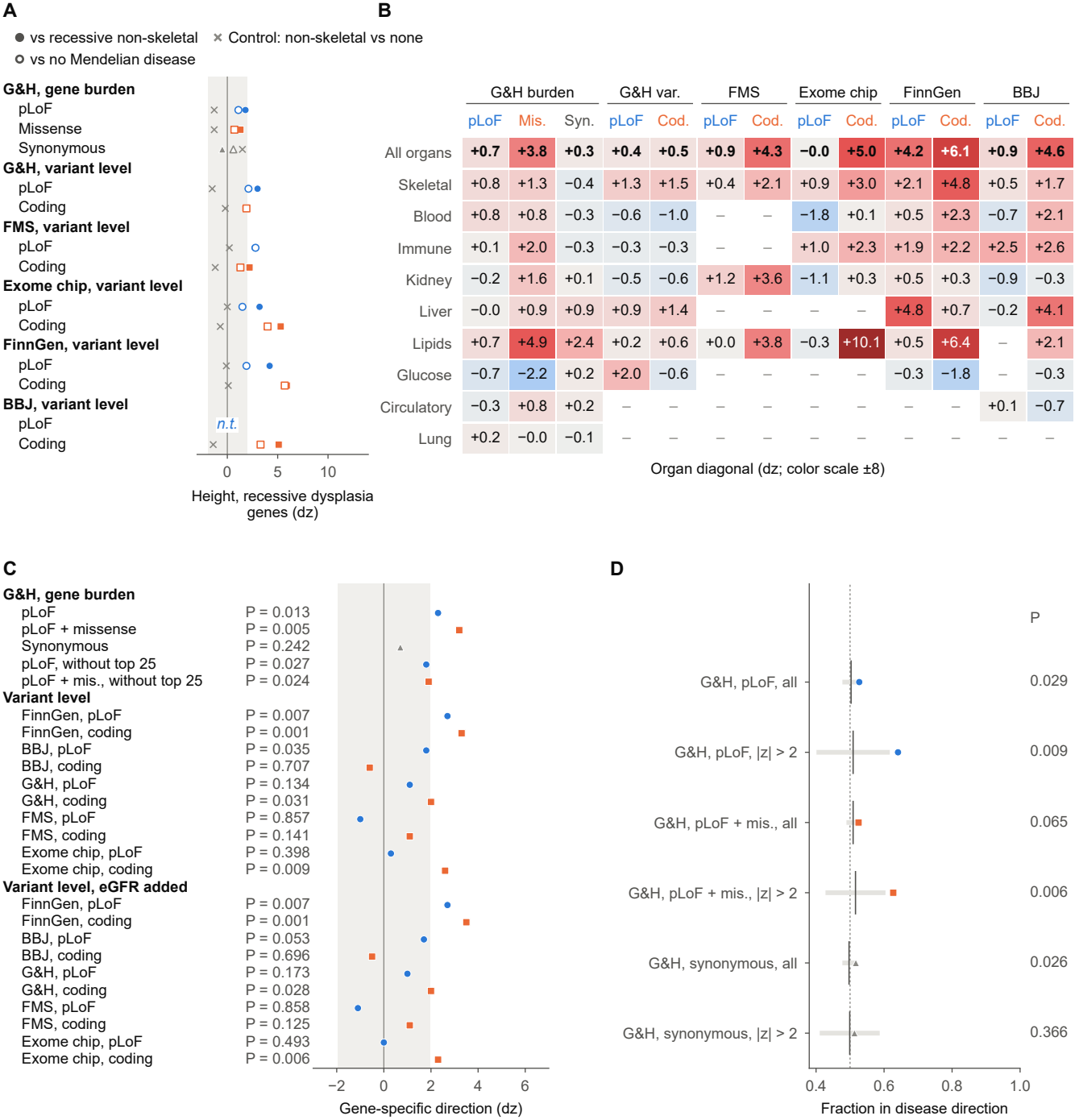

A
vs recessive non-skeletal
Control: non-skeletal vs none
vs no Mendelian disease
G&H, gene burden
pLoF
Missense
Synonymous
G&H, variant level
pLoF
Coding
FMS, variant level
pLoF
Coding
Exome chip, variant level
pLoF
Coding
FinnGen, variant level
pLoF
Coding
BBJ, variant level
pLoF
Coding
n.t.
0
5
10
Height, recessive dysplasia genes (dz)
B
G&H burden
G&H var.
FMS
Exome chip
FinnGen
BBJ
pLoF Mis. Syn. pLoF Cod. pLoF Cod. pLoF Cod. pLoF Cod. pLoF Cod.
All organs +0.7 +3.8 +0.3 +0.4 +0.5 +0.9 +4.3 −0.0 +5.0 +4.2 +6.1 +0.9 +4.6
Skeletal +0.8 +1.3 −0.4 +1.3 +1.5 +0.4 +2.1 +0.9 +3.0 +2.1 +4.8 +0.5 +1.7
Blood +0.8 +0.8 −0.3 −0.6 −1.0 – – −1.8 +0.1 +0.5 +2.3 −0.7 +2.1
Immune +0.1 +2.0 −0.3 −0.3 −0.3 – – +1.0 +2.3 +1.9 +2.2 +2.5 +2.6
Kidney −0.2 +1.6 +0.1 −0.5 −0.6 +1.2 +3.6 −1.1 +0.3 +0.5 +0.3 −0.9 −0.3
Liver −0.0 +0.9 +0.9 +0.9 +1.4 – – – – +4.8 +0.7 −0.2 +4.1
Lipids +0.7 +4.9 +2.4 +0.2 +0.6 +0.0 +3.8 −0.3 +10.1 +0.5 +6.4 – +2.1
Glucose −0.7 −2.2 +0.2 +2.0 −0.6 – – – – −0.3 −1.8 – −0.3
Circulatory −0.3 +0.8 +0.2 – – – – – – – – +0.1 −0.7
Lung +0.2 −0.0 −0.1 – – – – – – – – – –
Organ diagonal (dz; color scale ±8)
C
G&H, gene burden
pLoF P = 0.013
pLoF + missense P = 0.005
Synonymous P = 0.242
pLoF, without top 25 P = 0.027
pLoF + mis., without top 25 P = 0.024
Variant level
FinnGen, pLoF P = 0.007
FinnGen, coding P = 0.001
BBJ, pLoF P = 0.035
BBJ, coding P = 0.707
G&H, pLoF P = 0.134
G&H, coding P = 0.031
FMS, pLoF P = 0.857
FMS, coding P = 0.141
Exome chip, pLoF P = 0.398
Exome chip, coding P = 0.009
Variant level, eGFR added
FinnGen, pLoF P = 0.007
FinnGen, coding P = 0.001
BBJ, pLoF P = 0.053
BBJ, coding P = 0.696
G&H, pLoF P = 0.173
G&H, coding P = 0.028
FMS, pLoF P = 0.858
FMS, coding P = 0.125
Exome chip, pLoF P = 0.493
Exome chip, coding P = 0.006
−2
0
2
4
6
Gene-specific direction (dz)
D
P
G&H, pLoF, all 0.029
G&H, pLoF, |z| > 2 0.009
G&H, pLoF + mis., all 0.065
G&H, pLoF + mis., |z| > 2 0.006
G&H, synonymous, all 0.026
G&H, synonymous, |z| > 2 0.366
0.4
0.6
0.8
1.0
Fraction in disease direction